MIL-T-36245 A
5 May 1972
SUPERSEDING
MIL-T-36245(DM)
22 July 1964

MILITARY SPECIFICATION

TEST KIT, LACTIC DEHYDROGENASE ACTIVITY, 10 DETERMINATIONS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 The specification covers a Test Kit, Lactic Dehydrogenase Activity, Pyruvate Substrate, 10 Determinations, suitable for the accurate, colorimetric determination of lactic dehydrogenase in serum and body fluids. (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

PPP-C-00186a Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specification and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a Lactic Dehydrogenase Activity Test Kit prepared by the methods of Cabaud and Wroblewski. (Cabaud, P. G., Wroblewski F., Colorimetric Measurement of Lactic Dehydrogenase Activity of Body Fluids, American Journal Clinical Pathology, 30, 234, 1958), and suitable for 10 determinations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

3.1.1 Components. The test kit shall consist of the following:

Reagent No. 2: Lactic 2 Dehydrogenase: One (1) bottle of color developer, 16 ml.

Reagent No. 5: Lactic 5 Dehydrogenase: One (1) bottle of Buffered Pyruvate substrate, 16 ml.

Reagent No. 6: Lactic 6 Dehydrogenase: Ten (10) bottles each containing 1 mg Dihydrodiphospho-pyridine Nucleotide (DPNH).

3.1.2 Characteristics of each reagent.

3.1.2.1 Color developer (Reagent No. 2). The color developer (Reagent No. 2) shall be a light yellow, clear solution containing 2, 4 Dinitrophenylhydrazine in an approximately 1.0 N hydrochloric acid solution. The reagent shall contain no preservative.

3.1.2.2 Buffered pyruvate substrate (Reagent No. 5). The buffered pyruvate substrate (Reagent No. 5) shall be a colorless, clear solution containing sodium pyruvate buffered to a pH range between 7.3 and 7.5. The solution shall contain chloroform or an equivalent preservative as an antibacterial agent.

3.1.2.3 Dihydrodiphosphopyridine nucleotide (Reagent No. 6). The dihydrodiphosphopyridine nucleotide (Reagent No. 6) shall be dried and packaged preweighed in evacuated, reaction bottles (vials). Each vial shall contain 1 mg (-0, +0.2 mg) of reagent. The pre-weighed packaged reaction vials shall comply with the following:

When the contents of one (1) vial is diluted with 10 ml of purified water, the solution shall produce an optical density of not less than 0.5 at 340 mu through a 1 cm light path against an air blank.

3.1.2.4 Suitability of reagents. The suitability of the reagents shall be determined by performing the LDH assay using the colorimetric method described in the manufacturer's instructions on sera of known LDH activity. Multiple assays of the same serum shall agree within ± 5 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

3.1.3 Performance. The test kit, when used for the determination of LDH activity on sera according to the manufacturer's directions (colorimetric method), shall produce activity values that do not differ by more than ± 5 percent from the values obtained on the same, fresh, patient serum sample (not dried reference controls), using the standard reference ultraviolet method (Wroblewski and LaDue ultraviolet method, standardized to 25° C.)

3.2 Ingredients. All ingredients used in the manufacture of the reagents shall be of U.S.P. or N.F. quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade suitable for its intended use.

3.3 Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the product, to the date of delivery to the Government.

3.4 Workmanship. Workmanship shall be first class throughout. The material and its containers shall be free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:16Z
Check the source to verify that this is the current version before use.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of reagent produced from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.3 Inspection. Inspection shall be conducted as necessary to determine compliance with the requirements of this specification.

4.4 Sampling. Sampling shall be conducted in accordance with the procedures set forth in MIL-STD-105, with an acceptable quality level (AQL) of 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5 Tests.

4.5.1 Tests shall be conducted in accordance with the requirements set forth in this document.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of Interim Federal Specification PPP-C-00186a, dated 15 May 1969, and Amendment-1, dated 27 October 1969, and as specified herein:

5.1.1 Immediate containers. Each immediate container for the reagents shall comply with the following classification, as indicated for the reagent:

For Reagents No. 2 and No. 5:

GROUP A	CLASS 1	TYPE a or c	STYLE 1	GRADE 1

CLOSURE G

For Reagent No. 6:

GROUP A	CLASS 1	TYPE a or c	STYLE 1	GRADE 1

CLOSURE H

The immediate containers for Reagents No. 2, No. 5, and No. 6, shall be identical in size and design.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

5.2 Labeling. Labeling shall be in accordance with the applicable regulations, and shall include the information required below:

5.2.1 Immediate containers.

5.2.1.1 Reagent No. 2. Each immediate container of Reagent No. 2 shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) the commercial reagent name

(b) the designations "Reagent No. 2"
"COLOR DEVELOPER"
"A solution of 2, 4, DINITROPHENYLHYDRAZINE"

(c) the quantity of contents designated as "16 ml"

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the statement "Store between 2° - 8° C. (35° - 46° F.)"

(g) the date of manufacture.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:16Z
Check the source to verify that this is the current version before use.

5.2.1.2 Reagent No. 5. Each immediate container of Reagent No. 5 shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) the commercial reagent name

(b) the designations "Reagent No. 5" and "PYRUVATE SUBSTRATE"

(c) the quantity of contents designated as "16 ml"

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the statement "Store between 2° - 8° C. (35° - 46° F.)"

(g) the date of manufacture

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

5.2.1.3 Reagent No. 6. Each immediate container of Reagent No. 6 shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) the commercial reagent name

(b) the designations "Reagent No. 6" and "DIHYDRODIPHOSPHOPYRIDINE NUCLEOTIDE (DPNH)"

(c) the quantity of contents designated as "1 mg"

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the statement "Store between 2° - 8° C. (35° - 46° F.)"

(g) the date of manufacture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

5.2.2 Unit packages. Each unit package label shall bear the following information. However, the information is not required to appear in the sequence indicated:

(a) the item identification designated as "TEST KIT, LACTIC DEHYDROGENASE ACTIVITY, 10 Determinations"

(b) the designation "Pyruvate Substrate"

(c) the Federal Stock No. designated as "FSN 6505-890-1385" or "Stock No. 6505-890-1385"

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the statement "Store between 2° - 8° C. (35° - 46° F.)"

(g) the date of manufacture

(h) the statement "See enclosed pamphlet" or a similar statement.

5.2.3 Circulars. A circular, brochure, or pamphlet describing the items and their use shall be packaged within each unit package.

5.3 Packaging and packing.

5.3.1 Unit of issue. One test kit, as specified, constitutes one unit of issue.

5.3.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-00186a.

5.3.3 Procedure code No. 6 of PPP-C-00186a applies.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:16Z
Check the source to verify that this is the current version before use.

5.4 Marking.

5.4.1 Intermediate package. In paragraph 5.5.3 of PPP-C-00186a, add: "The date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"STORE BETWEEN 2° - 8° C. (35° - 46° F.)."

5.4.2 Exterior container. In paragraph 5.5.4 of PPP-C-00186a, add: "The date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"STORE BETWEEN 2° - 8° C. (35° - 46° F.)."

5.5 Special labeling.

5.5.1 For refrigerated shipment. Each refrigerated exterior container shall be marked as specified in MIL-STD-129.

5.5.2 For unrefrigerated shipment. Each unrefrigerated exterior container shall be marked as specified in MIL-STD-129.

5.5.3 The unrefrigerated shipping time for this item shall be not more than 168 hours.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement document should specify the following:

(a) Number, title, and date of this specification
(b) Federal Stock No. (FSN)
(d) Selection of applicable levels of packaging and packing (See PPP-C-00186a)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

6.2 Federal Stock No. coverage. This specification covers the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6505-890-1385	TEST KIT, LACTIC DEHYDROGENASE ACTIVITY, 10 Determinations

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1286

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:16Z
Check the source to verify that this is the current version before use.